Date: 18/03/2026

EX-7
Secure Network Architecture

AIM:

To design and implement a secure network architecture by applying security best practices to protect systems, data, and communication from unauthorized access and cyber threats.

PROCEDURE:

- Analyze network requirements and define security objectives.
- Segment the network into secure zones (LAN, DMZ, WAN) using VLANs.
- Configure firewall rules and access control lists (ACLs) to enforce traffic policies.
- Use secure protocols and encryption to protect data in transit.
- Monitor network activity continuously and update security configurations regularly.

TASK-1:

network and access important information. A system administrator should be confident that their servers are secure if a random device joins a network, knowing that the device is segmented from the rest of the network and cannot access those systems.

All of these concepts and scenarios are what separate a functional network from a well-designed network.

Throughout this room, we will break down these scenarios or objectives and understand different concepts we can use to implement them in a network. We will also discuss potential threats a network may face even after proper best practices are established.

Learning Objectives

1. Understand the principles of secure network architecture design
2. Learn and implement common network security concepts and protocols
3. Understand a network's environment and potential threats

Before beginning this room, we recommend that you have a basic understanding of common networking terminology and concepts.

Throughout this room we will introduce several commands and applications for demonstration purposes only. All questions can be answered using the content, images, and other provided materials.

Answer the questions below

Read the above and continue to the next task.

No answer needed | ✓ Correct Answer

TASK-2:

```
vyos@vyos-rtr# set interfaces ethernet eth0 vif 10 description 'VLAN 10'
vyos@vyos-rtr# set interfaces ethernet eth0 vif 10 address '192.168.100.1/24'
```

If all went well and was appropriately configured, we should be able to route traffic between VLANs while keeping traffic tagged and isolated!

But are they really isolated? Physically, they are isolated, but because routes exist between them, there is no security boundary, and they are not necessarily isolated. As long as a route exists between two VLANs, any device can communicate between the two.

This brings us to our following two tasks, where we will discuss designing secure VLANs and introduce the concept of zoning.

Analyzing a VLAN Configuration

Apply what you learned to analyze a VLAN configuration given the output below.

▶ Interface Configuration Snippet (Click to view)

Answer the questions below

How many trunks are present in this configuration?

4 | ✓ Correct Answer

What is the VLAN tag ID for interface eth12?

30 | ✓ Correct Answer

TASK-3:

Security zones and access controls will physically direct how and where traffic goes. But how is it decided what resources users or devices have access to? Traffic rules are often governed by company security policy or compliance as equally as security controls that determine access permissions.

We've now established a system to approach designing VLANs, but how can we practically implement and enforce them? In the next task, we will cover several protocols and applications that can be used to implement and enforce VLANs.

Answer the questions below

From the above table, what zone would a user connecting to a public web server be in?

External ✓ Correct Answer

From the above table, what zone would a public web server be in?

DMZ ✓ Correct Answer

From the above table, what zone would a core domain controller be placed in?

Restricted ✓ Correct Answer

TASK-4:

Analyzing Packets and ACLs

Now that we understand the structure of an ACL and what it will look for in a packet, let's analyze a few packets and ACL policies to determine if they will be accepted or dropped.

Below is each packet and ACL policy required; answer the questions using the resources below.

- Packet #1 (Click to view)
- ACL Policy #1 (Click to view)
- Packet #2 (Click to view)
- ACL Policy #2 (Click to view)

Answer the questions below

According to the corresponding ACL policy, will the first packet result in a drop or accept?

accept ✓ Correct Answer

According to the corresponding ACL policy, will the second packet result in a drop or accept?

drop ✓ Correct Answer

TASK-5:

DMZ	LAN	HTTP	Accept	Drop
DMZ	WAN	-	-	Drop
LAN	DMZ	ICMP	Accept	Drop
WAN	DMZ	-	-	Drop

Common Name	Interface	Default Action
DMZ	eth0.10	Drop
LAN	eth0.20	Drop
WAN	eth0.30	Drop

Answer the questions below

What is the flag found after filling in all blanks on the static site?

THM{M05tly_53cure}

✓ Correct Answer

TASK-6:

detected and monitored? If their beacon is using HTTPS through the DMZ. That will look like primarily legitimate traffic to your firewall and analysts.

To solve this issue, we must use SSL/TLS inspection to intercept HTTPS connections.

SSL/TLS Inspection

SSL/TLS inspection uses an **SSL proxy** to intercept protocols, including HTTP, POP3, SMTP, or other SSL/TLS encrypted traffic. Once intercepted, the proxy will decrypt the traffic and send it to be processed by a **UTM** (**U**nified **T**hreat **M**anagement) platform. UTM solutions will employ deep SSL inspection, feeding the decrypted traffic from the proxy into other UTM services, including but not limited to web filters or **IPS** (**I**ntrusion **P**revention **S**ystem), to process the information.

This solution may seem ideal, but what are the downsides? Some of you may have already noted that this requires an SSL proxy or **MitM** (**M**an-**i**n-**t**he-**M**iddle). Even if a firewall or vendor has already implemented the solution, it will still act as a MiTM between your devices and the outside world; what if it intercepts potentially plain-text passwords? A corporation must assess the pros and cons of this solution, dependent on its calculated risk. You could allow all applications that you know are safer to prevent potential cons, but this solution will still have disadvantages. For example, an advanced threat actor could route their traffic through a cloud provider or a trusted domain.

Answer the questions below

Does SSL inspection require a man-in-the-middle proxy? (Y/N)

Y

✓ Correct Answer

What platform processes data sent from an SSL proxy?

Unified Threat Management

✓ Correct Answer

TASK-7:

```
Router# show ip dhcp snoop bind
MacAddress         IpAddress       Lease(sec) Type          VLAN Interface
------------------ --------------- ---------- ------------- ---- --------------------
02:c8:85:b5:5a:aa  10.10.0.1       23453      dhcp-snooping 10   GigabitEthernet1/1
01:02:03:04:05:06  2.2.2.2         69445      dhcp-snooping 20   GigabitEthernet2/1
```

Answer the questions below

Where does DHCP snooping store leased IP addresses from untrusted hosts?

DHCP Binding Database — ✓ Correct Answer

Will a switch drop or accept a DHCPRELEASE packet?

Drop — ✓ Correct Answer

Does dynamic ARP inspection use the DHCP binding database? (Y/N)

Y — ✓ Correct Answer

Dynamic ARP inspection will match an IP address and what other packet detail?

MAC Address — ✓ Correct Answer

TASK-8:

Task 8 ✓ Conclusion

In this room, we have expanded the common requirements of a well-designed network to include security standards.

We have discussed approaches and solutions for problems commonly encountered at layers two and three of the OSI model. It may be apparent that this does not solve all issues, and each environment may have different requirements for what it should monitor and block.

To continue exploring network security beyond design, complete the Network Security Protocols room. You will go beyond the layers we discussed and observe security from layers seven to three that we did not discuss in this room.

Answer the questions below

Read the above and continue learning!

No answer needed — ✓ Correct Answer

RESULT:

A secure network architecture was successfully designed and implemented using VLANs, security zones (LAN, DMZ, WAN), ACL policies, SSL/TLS inspection, and DHCP snooping. Network traffic was effectively controlled and filtered based on defined rules. The experiment demonstrated that a well-structured secure network environment ensures controlled access, protected data transmission, and significantly reduced vulnerability to cyber attacks.